

Mobiles and Fisherman in Kerala, India

Before the arrival of mobile phones, fishermen in Kerala would land their catch at a particular market (there were several different markets along the coast), and try to get the best price (Jensen 2007). But if other fishermen had chosen the same market or there were few buyers at the market, they might get a poor price and also have to throw away unsold fish. Their main problem was information absence: they did not know prices and demand at other markets.

With mobile phones, the fishermen would call ahead to different fish markets along the coast and, as a result, they saw their earnings increase by an average Rs.205 (c.US\$4.5) per day. Most people assume this was because the fishermen were getting a higher price for their fish but that was not so: the average price per kg of fish actually fell. The increased revenue arose because the fishermen identified markets with higher demand, and so they sold more and wasted less of their catch. Less waste meant a higher supply of fish and, as per the laws of supply and demand, the price fell.

This digging-behind-the-headline is one reason why Jensen's work is an exemplary illustration of development informatics research, and he provides a number of other valuable insights:

- A classic ICT4D mistake is to just look at the benefits and forget the costs. Jensen calculates that fishermen were spending an average Rs.72 per day on mobile costs. So while gross revenue was up by Rs.205, net income was Rs.133 (c.US\$3) per day. Even this is faulty because it does not take into account the cost of the mobile phone – around Rs.5,000 (c.US\$110) at the time – which would consume around two months of additional income (given roughly 24 days per month of fishing).
- Alongside benefits for the phone owners, there was also a digital provide to others. “Those fishermen without mobile phones also saw their profit rise by an average Rs.97 (c.US\$2) per day as a result of the general improvements in market efficiency which phones introduced. This was about half the profit increase seen by phone owners and meant, even allowing for the additional costs, that returns to phone ownership were greater than those for non-ownership” (Heeks 2014d:14). And, because lower wastage and higher supply reduced the price of fish, consumers also benefited.
- Fishermen also benefited from use of mobiles by others; for example, they lost less fishing time due to equipment failures when repair companies adopted mobiles, and thus sped up their transaction times (Jensen 2010).

This is therefore a very positive ICT4D story – a win-win for all those involved – which may be one reason it has been so widely circulated and widely used.

However, this isn't the whole story. A re-investigation some years later found that – while the overall results of Jensen's study held – there were additional issues (Srinivasan & Burrell 2015). All fishermen could now afford phones so the digital provide to non-phone-owners had disappeared. The design assumption that fishermen choose between different markets to land their catch was true for larger boats, but was rarely true in reality for small boats, which typically used the same market every day. So the impact of phones was uneven across different fishermen. And phones were valued as much for other tasks such as rescue at sea and fish-finding (in conjunction with GPS (global positioning system) location capabilities) as they were for help with pricing.

